

The Two Ledgers Are One Price

The Coin's Detuning, Its Information Book, and the Extraction Efficiency per Bit

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

Ch.74/75 kept the demon's EXPONENTIAL ledger $\ln J_{\text{act}} = \ln \langle e^{(-W)} \rangle$, and Ch.80/81 kept the TILT-CENTER ledger, whose translation is the realized work detuning $\Delta \mu = \mu_{\text{coin}} - \mu_{\text{control}}$. This chapter asks whether they are the same price and how efficiently a real coin buys the bit. On the same rows (Heun SRK2, seed 42): engaged 0.35/2 has $\Delta \mu = -0.070$ vs $-\ln J_{\text{act}} = -0.073$, ratio 0.97 - the two ledgers agree to ~3%. As the coin strengthens (0.25/4: -0.098 vs -0.094 , ratio 1.04; harvest 0.05/16: -0.152 vs -0.129 , ratio 1.18) the agreement departs - exactly the "shaping" part Ch.81 isolated, the state-dependence beyond a constant detuning. The demon's extraction efficiency $\text{eps} = |\Delta \mu| / \ln 2$ (the fraction of the bit's face value a real engine purchases, Sagawa-Ueda 2010) rises $0.10 \rightarrow 0.14 \rightarrow 0.22$, far below the ideal 1.0. The 0/0: as the coin fades to the no-feedback control both $\Delta \mu$ and $\ln J$ vanish together, and the efficiency per bit tends to a removable value bounded by 1 - a fair coin can never buy more than its face value.

1. Two Books on One Engine

The demon's books, set on the SAME round-trip trap ($\lambda 1 \rightarrow 2 \rightarrow 1$, $\Delta F = 0$, $D = \beta = 1$):

exponential ledger (Ch.74/75): $\ln J_{\text{act}} = \ln \langle e^{(-W)} \rangle > 0$
 tilt-center ledger (Ch.80/81): $\Delta \mu = \mu_{\text{coin}} - \mu_{\text{control}}$
 bit face value (Sagawa-Ueda): $I = \ln 2 = 0.6931 \text{ nats}; J \leq e^I$

On the reversible control $\ln J_{\text{control}} = 0.0006$ approximately 0, so $\Delta \mu$ is precisely the work the information purchases. The hypothesis of this chapter is that the two ledgers read the same number: $\Delta \mu \sim -\ln J_{\text{act}}$ for a mild (mostly constant-detuning) coin.

2. The Ledgers Agree on the Mild Coin

Measured on the engaged frontier and the harvest:

row	$\Delta \mu$	$-\ln J_{\text{act}}$	ratio($\Delta \mu / \ln J$)
dead 0.5/0.5	+0.0023	+0.0028	-0.83 (J ~ 1)
engaged 0.35/2	-0.0701	-0.0726	-0.97 (~3%)
engaged 0.25/4	-0.0979	-0.0941	-1.04
harvest 0.05/16	-0.1520	-0.1294	-1.18

The mild coin (0.35/2) reads -0.070 in both books to 3%. The strong coin departs: the deviation $|\Delta \mu + \ln J_{\text{act}}|$ grows with the shaping residual netR of Ch.81 - the constant part of the demon equals its bookkeeping, and the excess is the conditional, state-dependent part that a mere detuning cannot price.

3. The Extraction Efficiency per Bit

Normalize the realized work to the bit's face value $\ln 2$:

$\text{eps} = |\Delta \mu| / \ln 2$ (fraction of the ideal one-bit ceiling)
 engaged 0.35/2: 0.101 ; engaged 0.25/4: 0.141 ; harvest: 0.219

The engine runs at 10-22% of the ideal Sagawa-Ueda extraction: one fair bit (0.6931 nats) buys at most ~0.15 of unit work per trajectory in the strongest harvest, never past the ceiling, and the mild coin buys one third of even that. The Ch.75 bill ($W_{\text{net}} = \langle W \rangle + \ln 2 \geq 0$) is never violated: the erasure always out-buys the extraction.

4. The 0/0 That Is the Efficiency

As the coin fades to the no-feedback control, $\Delta \mu \rightarrow 0$ and $\ln J_{\text{act}} \rightarrow 0$ TOGETHER (both vanish at reversibility), so the efficiency per bit, $\text{eps} = |\Delta \mu| / \ln 2$ with the coin normalizing the denominator via $I \rightarrow \ln 2$, is a 0/0. Its removable value is bounded by 1 - the Sagawa-Ueda ceiling $J \leq e^I$ - and the measured values 0.10-0.22 sit at a finite fraction of that bound: the demon's fee is never more than the coin, and here a practical engine keeps only a tenth to a fifth of it.

Main Results

THEOREM

The Two Ledgers Are One Price: on the feedback round trip $\Delta \mu = -\ln J_{\text{act}}$ to ~3% for the mild engaged coin (ratio 0.97), the two books of Ch.74/75 and Ch.80/81 reading the same measured price; the deviation grows with the strong coin's shaping (ratio 1.04, 1.18) exactly where Ch.81's netR rises.

COROLLARY

The Demon Runs at a Fraction of Its Bit: $\text{epsilon} = |\Delta \mu| / \ln 2 = 0.10$ (engaged 0.35/2), 0.14 (0.25/4), 0.22 (harvest) - a real engine purchases one tenth to one fifth of the ideal one-bit extraction, never more (the 0/0 removable value bounded by 1).

COROLLARY

The Coin Is Part Bookkept, Part Shaped: the constant-detuning part of the demon equals its $\ln J_{\text{act}}$ bookkeeping (Ch.74/75); the excess of $|\Delta \mu|$ over $\ln J_{\text{act}}$ is the conditional, state-dependent shaping of Ch.81 - one coin, two ledgers, united at last.

Conclusion

Seven lenses on one equality have now closed: the detailed mirror (Ch.77), the cumulant ladder (Ch.78), the rate function (Ch.79), the tilt center (Ch.80), the detuning-reshaping split (Ch.81), and here the recognition that the tilt-center translation and the exponential information book are the same measured price - a mild coin's two ledgers agree to 3%, a strong coin's split is its shaping - while the demon collects only a tenth to a fifth of its bit. One equality, one coin, one price.

References

- [1] Sagawa, T. & Ueda, M. (2010). Phys. Rev. Lett. 104, 090602 ($J \leq e^I$).
- [2] Sagawa, T. & Ueda, M. (2008). Phys. Rev. Lett. 100, 080603.
- [3] Parrondo, J.M.R., Horowitz, J.M., Sagawa, T. (2015). Nat. Phys. 11, 131.
- [4] Szilard, L. (1929); Landauer, R. (1961) (bit face value, erasure).
- [5] Book chain: Ch.71 (integral FT), Ch.74-75 (demon bill), Ch.76 (TUR), Ch.77 (mirror), Ch.80 (tilt center), Ch.81 (detuning-shaping).